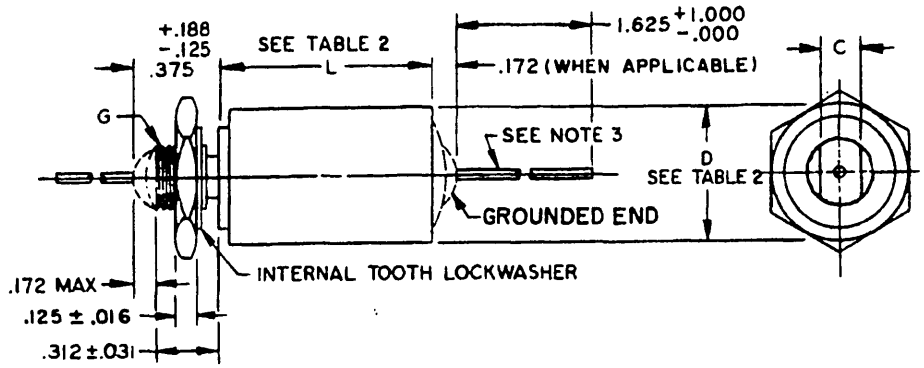
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, THREADED-STUD RETAINER, TUBULAR (UNINSULATED) (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED RELIABILITY, STYLE CQR13

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-19978.



INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.125	3.18	.437	11.10
.004	.10	.172	4.37	.562	14.27
.005	.13	.188	4.78	.670	17.02
.016	.41	.250	6.35	.750	19.05
.031	.79	.312	7.92	1.000	25.40
.032	.81	.375	9.53	1.625	41.28

Case dimension D	C ± .005	G
.562 or less	.250	5/16-24UNF-2A
.670	.375	7/16-28UNEF-2A
.750 and up	.437	1/2-28UNEF-2A

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Leads shall be of tinned solidwire , .032 +.004, -.001 (No. 20 AWG).
4. Mounting nuts and lockwashers shall be supplied.

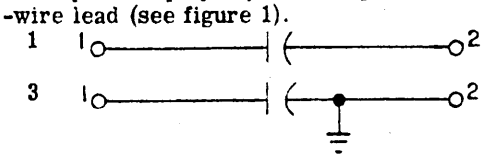
FIGURE 1. Style CQR13 capacitors.

FSC 5910

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REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1 and table II.
Case material - Nonmagnetic (screwneck and end seal may be of magnetic material).
Dielectric material - Paper and polyethylene terephthalate(K).
Terminals - Axial-wire lead (see figure 1).
Circuit diagram -



Rated voltage - See table II.
Rated temperature - -65° to +125°C.
Capacitance (cap.) (nom) - See table II.
Capacitance tolerance - See table II.
Dissipation factor (DF) (max) - 1.0 percent.
Failure rate level - In accordance with MIL-C-19978.
Reliability - In accordance with MIL-C-19978.

Burn-in: In accordance with MIL-C-19978.

Radiographic inspection: In accordance with MIL-C-19978.

Seal: Method 112 of MIL-STD-202, test condition letter - A.

Barometric pressure: In accordance with MIL-C-19978.
Test points:
Circuit diagram 1 - Between terminals and case.
Circuit diagram 3 - Between ungrounded terminal and case.

Insulation resistance (IR):
Terminal to terminal - See table I.

TABLE I. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminal to case - Greater than 10,000 megohms.

Vibration high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:
Direction and duration of motion - 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-C-19978.

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TABLE II. Style CQR13 capacitors.

Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate levels	Case dimensions			
					Circuit 1		Circuit 3	
					L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF						
CQR13A-KC392-3 -	200	.0039	G, J, K	L, M, P, R, S	.750	.400	.688	.400
CQR13A-KC472-3 -	200	.0047	G, J, K		.750	.400	.688	.400
CQR13A-KC562-3 -	200	.0056	G, J, K		.750	.400	.688	.400
CQR13A-KC682-3 -	200	.0068	G, J, K		.750	.400	.688	.400
CQR13A-KC183-3 -	200	.018	G, J, K		.875	.400	.812	.400
CQR13A-KC223-3 -	200	.022	G, J, K		.875	.400	.812	.400
CQR13A-KC273-3 -	200	.027	G, J, K		.875	.400	.812	.400
CQR13A-KC333-3 -	200	.033	G, J, K		.875	.400	.812	.400
CQR13A-KC393-3 -	200	.039	G, J, K		.875	.400	.812	.400
CQR13A-KC473-3 -	200	.047	G, J, K		.875	.400	.812	.400
CQR13A-KC563-3 -	200	.056	G, J, K		.875	.400	.812	.400
CQR13A-KC683-3 -	200	.068	G, J, K		.875	.400	.812	.400
CQR13A-KC823-3 -	200	.082	G, J, K		1.125	.400	1.062	.400
CQR13A-KC104-3 -	200	.10	G, J, K		1.125	.400	1.062	.400
CQR13A-KC124-3 -	200	.12	G, J, K		1.375	.400	1.312	.400
CQR13A-KC154-3 -	200	.15	G, J, K		1.375	.400	1.312	.400
CQR13A-KC184-3 -	200	.18	G, J, K		1.125	.562	1.062	.562
CQR13A-KC224-3 -	200	.22	G, J, K		1.125	.562	1.062	.562
CQR13A-KC274-3 -	200	.27	G, J, K		1.375	.562	1.312	.562
CQR13A-KC334-3 -	200	.33	G, J, K		1.375	.562	1.312	.562
CQR13A-KC394-3 -	200	.39	G, J, K		1.625	.562	1.562	.562
CQR13A-KC474-3 -	200	.47	G, J, K		1.625	.562	1.562	.562
CQR13A-KC564-3 -	200	.56	G, J, K		1.625	.670	1.562	.670
CQR13A-KC684-3 -	200	.68	G, J, K		1.625	.670	1.562	.670
CQR13A-KC824-3 -	200	.82	G, J, K		2.125	.750	2.062	.750
CQR13A-KC105-3 -	200	1.000	G, J, K		2.125	.750	2.062	.750
CQR13A-KE272-3 -	400	.0027	G, J, K		.750	.400	.688	.400
CQR13A-KE332-3 -	400	.0033	G, J, K		.750	.400	.688	.400
CQR13A-KE123-3 -	400	.012	G, J, K		.875	.400	.812	.400
CQR13A-KE153-3 -	400	.015	G, J, K		.875	.400	.812	.400
CQR13A-KE273-3 -	400	.027	G, J, K		.875	.400	.812	.400
CQR13A-KE333-3 -	400	.033	G, J, K		.875	.400	.812	.400
CQR13A-KE393-3 -	400	.039	G, J, K		1.125	.400	1.062	.400
CQR13A-KE473-3 -	400	.047	G, J, K		1.125	.400	1.062	.400
CQR13A-KE563-3 -	400	.056	G, J, K		1.375	.400	1.312	.400
CQR13A-KE683-3 -	400	.068	G, J, K		1.375	.400	1.312	.400
CQR13A-KE823-3 -	400	.082	G, J, K		1.125	.562	1.062	.562
CQR13A-KE104-3 -	400	.10	G, J, K		1.125	.562	1.062	.562
CQR13A-KE124-3 -	400	.12	G, J, K		1.375	.562	1.312	.562
CQR13A-KE154-3 -	400	.15	G, J, K		1.375	.562	1.312	.562
CQR13A-KE184-3 -	400	.18	G, J, K		1.625	.562	1.562	.562
CQR13A-KE224-3 -	400	.22	G, J, K		1.625	.562	1.562	.562
CQR13A-KE274-3 -	400	.27	G, J, K		1.625	.670	1.562	.670
CQR13A-KE334-3 -	400	.33	G, J, K		1.625	.670	1.562	.670
CQR13A-KE394-3 -	400	.39	G, J, K		2.125	.750	2.062	.750
CQR13A-KE474-3 -	400	.47	G, J, K	L, M, P, R, S	2.125	.750	2.062	.750

See footnote at end of table.

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TABLE II. Style CQR13 capacitors. - Continued

Part number 1/	DC voltage rating	Capacitance rating	Capacitance tolerance	Failure rate levels	Case dimensions			
					Circuit 1		Circuit 3	
					L ±.031	D +.015 -.005	L ±.031	D +.015 -.005
	Volts	µF						
CQR13A-KF102-3	600	.0010	G, J, K	M, P, R, S	.750	.235	.688	.400
CQR13A-KF122-3	600	.0012	G, J, K		.750	.235	.688	.400
CQR13A-KF152-3	600	.0015	G, J, K		.750	.235	.688	.400
CQR13A-KF182-3	600	.0018	G, J, K		.750	.235	.688	.400
CQR13A-KF222-3	600	.0022	G, J, K		.750	.235	.688	.400
CQR13A-KF272-3	600	.0027	G, J, K		.875	.312	.812	.400
CQR13A-KF332-3	600	.0033	G, J, K		.875	.312	.812	.400
CQR13A-KF392-3	600	.0039	G, J, K		.875	.312	.812	.400
CQR13A-KF472-3	600	.0047	G, J, K		.875	.312	.812	.400
CQR13A-KF562-3	600	.0056	G, J, K		.875	.312	.812	.400
CQR13A-KF682-3	600	.0068	G, J, K		.875	.312	.812	.400
CQR13A-KF822-3	600	.0082	G, J, K		.875	.312	.812	.400
CQR13A-KF103-3	600	.010	G, J, K		.875	.312	.812	.400
CQR13A-KF123-3	600	.012	G, J, K		.875	.400	.812	.400
CQR13A-KF153-3	600	.015	G, J, K		.875	.400	.812	.400
CQR13A-KF183-3	600	.018	G, J, K		.875	.400	.812	.400
CQR13A-KF223-3	600	.022	G, J, K		.875	.400	.812	.400
CQR13A-KF273-3	600	.027	G, J, K		1.125	.400	1.062	.400
CQR13A-KF333-3	600	.033	G, J, K		1.125	.400	1.062	.400
CQR13A-KF393-3	600	.039	G, J, K		1.375	.400	1.312	.400
CQR13A-KF473-3	600	.047	G, J, K		1.375	.400	1.312	.400
CQR13A-KF563-3	600	.056	G, J, K		1.125	.562	1.062	.562
CQR13A-KF683-3	600	.068	G, J, K		1.125	.562	1.062	.562
CQR13A-KF823-3	600	.082	G, J, K		1.375	.562	1.312	.562
CQR13A-KF104-3	600	.10	G, J, K		1.375	.562	1.312	.562
CQR13A-KF124-3	600	.12	G, J, K		1.625	.562	1.562	.562
CQR13A-KF154-3	600	.15	G, J, K		1.625	.562	1.562	.562
CQR13A-KF184-3	600	.18	G, J, K		1.625	.670	1.562	.670
CQR13A-KF224-3	600	.22	G, J, K		1.625	.670	1.562	.670
CQR13A-KF274-3	600	.27	G, J, K		2.125	.750	2.062	.750
CQR13A-KF334-3	600	.33	G, J, K		2.125	.750	2.062	.750
CQR13A-KF394-3	600	.39	G, J, K		2.375	.750	2.312	.750
CQR13A-KF474-3	600	.47	G, J, K		2.375	.750	2.312	.750
CQR13A-KG184-3	1000	.18	G, J, K		2.125	.750	2.062	.750
CQR13A-KG224-3	1000	.22	G, J, K		2.125	.750	2.062	.750
CQR13A-KG274-3	1000	.27	G, J, K		2.125	1.0	2.062	1.000
CQR13A-KG334-3	1000	.33	G, J, K		2.125	1.0	2.062	1.000
CQR13A-KG394-3	1000	.39	G, J, K		2.375	1.0	2.312	1.000
CQR13A-KG474-3	1000	.47	G, J, K	M, P, R, S	2.375	1.0	2.312	1.000

1/ Complete part number shall include additional symbols to indicate circuit, capacitance tolerance, and failure rate level as applicable.

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Solderability: In accordance with MIL-C-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter - D.

Stability at low and high temperature:
Low temperature:
Test temperature - $-65^{\circ}\ {}^{+0^{\circ}}_{-5^{\circ}}$C.
Capacitance change (max) - -10 percent.
High temperature:
Test temperature - $+125^{\circ}\ {}^{+5^{\circ}}_{-0^{\circ}}$C
Capacitance change (max) - +10 percent.

Life:
Capacitance change (max) - ±5 percent of initial measured value.

Resistance to soldering heat: In accordance with MIL-C-19978.

TABLE III. Millimeter equivalents of decimal inches.

INCHES	MM	INCHES	MM	INCHES	MM
0.005	0.13	0.688	17.48	1.312	33.32
0.015	0.38	0.750	19.05	1.375	34.93
0.031	0.79	0.812	20.62	1.562	39.67
0.235	5.97	0.833	21.16	1.625	41.28
0.312	7.92	0.875	22.23	2.062	52.37
0.400	10.16	1.000	25.40	2.125	53.98
0.562	14.27	1.062	26.97	2.312	58.72
0.670	17.02	1.125	28.58	2.375	60.33

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI, MU
Navy - AS
Air Force - 11, 17
DSA - ES

User activities:
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Navy - MC, AS, OS, SH
Air Force -

Preparing activity:
Air Force - 80

Agent:
DSA - ES

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